



School of Engineering Technology

Main Campus, Off Hennur-Bagalur Main Road, Chalahatti, Bengaluru-562149

MINI PROJECT REPORT

on

“Supermarket Inventory Management System”

submitted to,

School of Engineering and Technology, CMR University

in partial fulfilment of the requirement for the award of the degree of

Bachelor of Technology,

in

COMPUTER SCIENCE AND ENGINEERING(AI&ML)

by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING-(AI&ML)

CMR UNIVERSITY

2025-26

School of Engineering and Technology, CMR

Main Campus, Off Hennur-Bagalur Main Road, Chagalahatti, Bengaluru-562149

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING-(AI&ML)

CERTIFICATE

Certified that the mini project titled **“Supermarket Inventory Management System”** carried out by Mr./Ms. **Prajwal kk (24BBTCA088)**, **Neysa mary Pramod (24BBTCA078)**, **Pooja Rajpurohit (24BBTCA084)** and **Kavya Shree V (24BBTCA060)** in partial fulfilment for the award of Bachelor of Technology in **COMPUTER SCIENCE AND ENGINEERING-(AI&ML)** of CMR University, during the year 2025-2026. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The mini project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

Examiners

Name of the examiners

1. Prof. Ranjitha R

Signature

Prof Ranjitha R

Assistant Professor
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Signature with date

DECLARATION

We, **Prajwal KK, Neysa Mary Pramod, Pooja Rajpurohit, Kavya Shree V** students of School of Engineering and Technology, CMR university, hereby declare that the dissertation titled “**Super Market Inventory Management System**” embodies the report of my mini project carried out independently by us during fifth semester of **Bachelor of Technology in Computer Science and Engineering(AI&MI)**, under the supervision of **Prof.Ranjitha R**, Department of Computer Science and Engineering and this work has been submitted in partial fulfilment for the award of the **Bachelor of Technology** degree.

We have not submitted the project for the award of any other degree of any other university or institution.

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Abstract

The rapid growth of retail businesses has increased the need for efficient and accurate inventory management systems. Traditional manual methods of maintaining stock and transaction records are time-consuming, error-prone, and inefficient. This project presents the design and implementation of a Supermarket Inventory Management System aimed at digitizing and automating the management of products, suppliers, customers, and sales transactions.

The primary objective of this system is to ensure real-time tracking of stock availability, reduce human errors, and improve overall operational efficiency.

The primary objective of this system is to provide an easy-to-use interface for maintaining accurate inventory records, including product name, price, and quantity. The system enables users to perform essential operations such as adding new products, updating stock quantities, selling products, deleting items, and searching for specific products using unique identifiers or names. It ensures real-time updates of inventory by automatically modifying stock levels during sales and restocking processes, thereby reducing inconsistencies and improving reliability.

The methodology involves designing a structured relational database using MySQL to store and manage product information efficiently. Backend functionality is implemented using PHP, which handles data processing and communication with the database, while the frontend is developed using HTML and CSS to provide a simple and interactive user interface. The system is built on CRUD (Create, Read, Update, Delete) operations, ensuring efficient data manipulation.

The results demonstrate that the system significantly improves data accuracy, reduces manual workload, and enables faster access to inventory information. Overall, it provides a practical and scalable solution for small and medium-sized supermarkets to manage their inventory effectively.

Keywords: Inventory Management, DBMS, PHP, MySQL, CRUD Operations, Supermarket System, Stock Management

1. Introduction

Background / Motivation of the Project

With the rapid growth of retail businesses and supermarkets, managing inventory efficiently has become a crucial task. Traditionally, supermarkets relied on manual methods such as handwritten records or basic spreadsheets to track products and stock levels. These methods are not only time-consuming but also prone to errors, data inconsistency, and difficulty in handling large volumes of information. The need for an automated and reliable system to manage inventory has motivated the development of this Supermarket Inventory Management System. The project aims to replace manual processes with a digital solution that is faster, more accurate, and easier to manage. The motivation behind this project is to design a system that simplifies inventory management by automating routine tasks such as stock updating, product tracking, and record maintenance. By using a database-driven approach, the system ensures that all product-related information is stored systematically and can be accessed easily whenever required.

Importance / Relevance of the Topic

Inventory management is a fundamental aspect of any retail business, as it directly impacts sales, customer satisfaction, and overall profitability. An efficient system ensures that products are always available when needed, prevents overstocking or understocking, and maintains accurate records of all items. In today's digital era, businesses are increasingly adopting database-driven systems to streamline their operations. This project is highly relevant as it demonstrates how database management systems (DBMS) can be applied in real-world scenarios to improve efficiency and reliability in supermarkets.

This project is highly relevant as it demonstrates the practical application of Database Management Systems (DBMS) concepts such as data storage, retrieval, and manipulation. It also reflects real-world scenarios where businesses rely on digital systems to improve productivity and decision-making.

Problem Overview

Manual inventory systems often lead to several challenges, including inaccurate stock records, difficulty in tracking product availability, delays in updating data, and increased chances of human error. Additionally, retrieving specific information from large datasets becomes cumbersome without a proper system. These issues can result in financial losses and poor decision-making. The absence of a centralized system makes it difficult to manage product data effectively.

The lack of a centralized system makes it difficult to maintain consistency and accuracy in data. Therefore, there is a need for an automated system that can efficiently manage product details, update stock in real time, and provide quick access to information.

Scope and Significance

The scope of this project is focused on developing a simple and efficient inventory management system for supermarkets. It includes functionalities such as adding products, updating stock, deleting records, searching for products, and managing sales operations.

The significance of this system lies in its ability to reduce manual effort, improve data accuracy, and enhance overall efficiency. It serves as a foundational model that can be further expanded with advanced features like supplier management, billing systems, and reporting tools.

Organization of the Report

This report is organized into multiple sections for better understanding of the project. The introduction provides an overview of the system and its importance. The subsequent sections describe the system design, database structure, and implementation details. Further sections explain the results and outcomes of the project, followed by advantages and applications. Finally, the report concludes with a summary of findings and possible future enhancements.

4. Problem Statement

A clear and concise statement describing:

In the modern retail industry, supermarkets handle a wide range of products and a continuous flow of customer transactions on a daily basis. Managing such a large volume of data using traditional methods such as manual record-keeping or basic spreadsheets often leads to inefficiencies, inconsistencies, and errors. Many small and medium-scale supermarkets still rely on these outdated approaches, which are not capable of handling dynamic inventory requirements effectively. This creates a significant need for a reliable and automated system that can manage inventory data accurately and efficiently.

What problem is being solved

One of the major problems faced in traditional inventory management systems is the lack of real-time stock tracking. When product quantities are updated manually, there is a high possibility of errors such as incorrect entries, missed updates, or duplication of data. These inaccuracies can lead to serious issues like overstocking, where excess products occupy storage space and may even expire, or understocking, where essential items are unavailable, resulting in customer dissatisfaction and loss of sales. Furthermore, manual systems make it difficult to quickly identify which products are available, which are running low, and which need to be restocked.

Why it is important

The importance of solving these issues lies in the direct impact inventory management has on business operations and customer satisfaction. An efficient inventory system ensures that products are available when needed, reduces unnecessary costs, and improves overall operational efficiency. In a competitive retail environment, even small inefficiencies can result in significant losses. Therefore, adopting a digital solution is not just beneficial but essential for sustainable growth.

Who benefits from the solution

This project aims to address these challenges by developing a Supermarket Inventory Management System that automates and simplifies inventory-related tasks. The system is designed to maintain a structured and centralized database of products, including details such as product name, price, and quantity. It provides functionalities to add new products, update stock levels, delete records, and search for specific items quickly and accurately. By implementing real-time updates, the system ensures that stock levels are always current, reducing the chances of errors and inconsistencies.

5.Objectives

The primary objective of this project is to design and develop a Supermarket Inventory Management System that enables efficient and systematic management of product records and stock levels in a digital environment. The system aims to replace traditional manual methods with an automated solution that improves accuracy, reliability, and ease of data handling. By digitizing inventory operations, the project ensures that all product-related information is stored in a structured and organized manner.

Another important objective is to design and implement a relational database using MySQL for efficient data storage and management. The database is structured to store essential product details such as product ID, name, price, and quantity, ensuring proper organization and easy retrieval of data. The use of a database management system helps maintain data integrity, reduce redundancy, and support efficient query processing.

The project also aims to implement CRUD (Create, Read, Update, Delete) operations, which form the core functionality of the system. These operations allow users to add new products, view available products, update stock quantities, and delete products when necessary. This ensures that the system can handle all essential inventory-related tasks efficiently and effectively.

A key objective of the system is to provide real-time tracking and updating of inventory. The system is designed in such a way that stock levels are automatically updated during product addition and sales operations. This helps maintain accurate and up-to-date inventory records, reducing the chances of errors such as overstocking or stock shortages.

In addition, the project focuses on developing a user-friendly and interactive interface using HTML and CSS. The interface is designed to be simple and easy to use, allowing users to perform operations without requiring advanced technical knowledge. Features such as forms, buttons, and tables are used to enhance usability and improve user experience.

Another objective is to implement an efficient search functionality that allows users to quickly retrieve product information based on product ID or name. This reduces the time required to locate specific records and improves the overall efficiency of the system.

Finally, the system is designed with scalability and flexibility in mind. It can be extended in the future to include advanced features such as billing systems, supplier management, sales reports, and analytics. This makes the project a strong foundation for developing a more comprehensive retail management system.

6. System Analysis

Existing System

In many small and medium-scale supermarkets, inventory management is still carried out using manual methods such as maintaining records in notebooks or basic spreadsheet tools. These traditional systems have several limitations and drawbacks.

Firstly, manual record-keeping is highly time-consuming and prone to human errors. Mistakes in data entry, miscalculations, or missed updates can lead to inaccurate stock records. Secondly, there is no real-time tracking of inventory, making it difficult to know the exact availability of products at any given time. This often results in issues such as overstocking or stock shortages.

Additionally, searching for specific product information in manual systems is inefficient and slow, especially when dealing with a large number of products. Data redundancy and inconsistency are also common problems due to the lack of a centralized database. Furthermore, maintaining and updating records requires significant effort, reducing overall productivity. These limitations highlight the need for a more efficient and automated system.

Proposed System

The proposed system is a Supermarket Inventory Management System designed to overcome the limitations of the existing manual methods. It is a web-based application developed using PHP, MySQL, HTML, and CSS, which provides a digital platform for managing inventory efficiently.

The system allows users to perform essential operations such as adding new products, updating stock quantities, selling products, deleting records, and searching for specific items. It maintains a centralized database where all product information is stored in an organized manner.

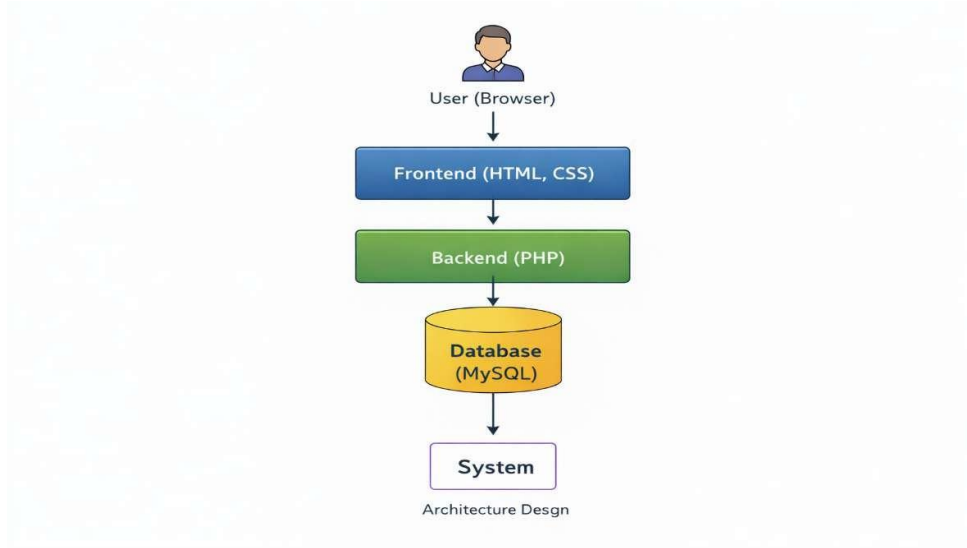
One of the major advantages of the proposed system is real-time stock updating, where inventory levels are automatically adjusted during product addition and sales. This ensures accuracy and reduces inconsistencies. The system also provides a user-friendly interface, making it easy to use even for non-technical users.

Overall, the proposed system improves efficiency, reduces manual workload, enhances data accuracy, and provides quick access to information, making it a reliable solution for supermarket inventory management.

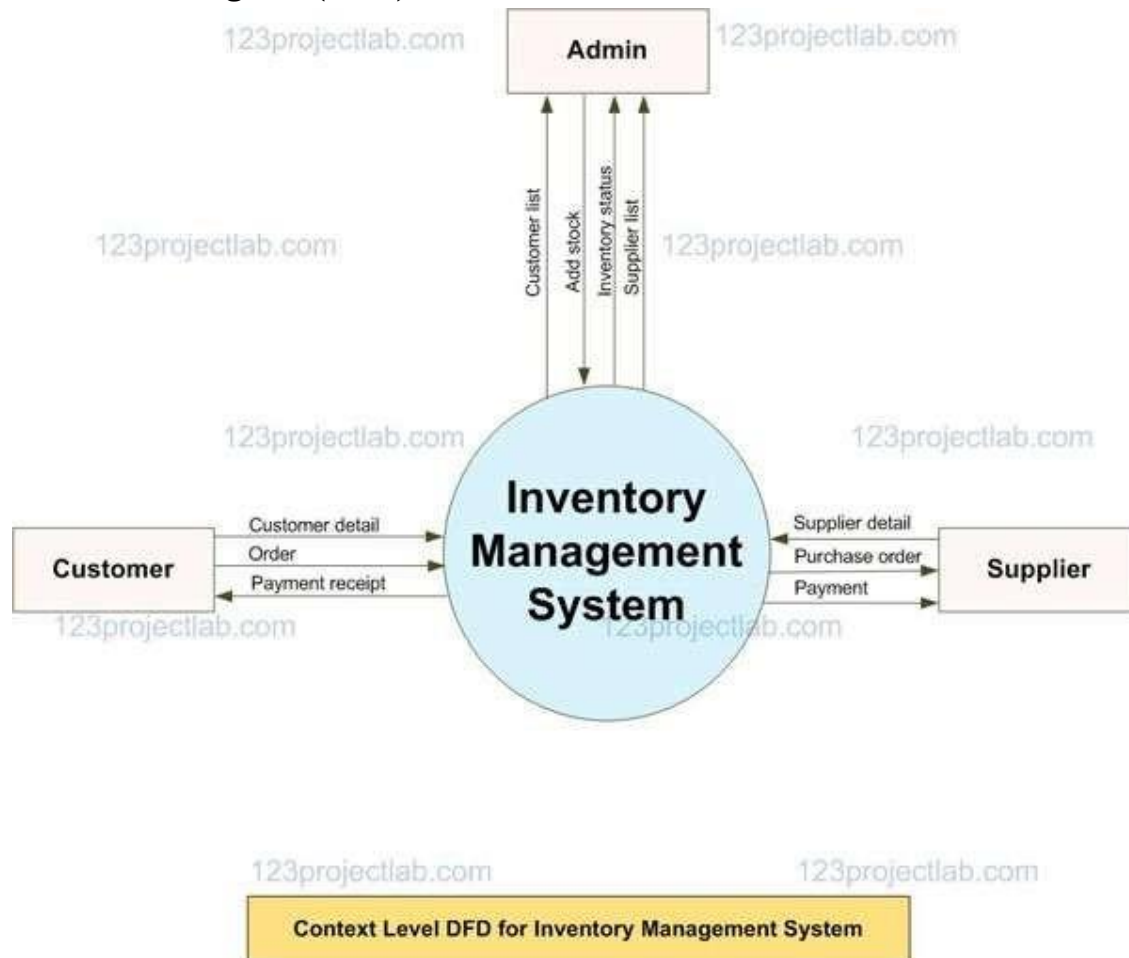
System Requirements			
Hardware Requirements	Specification	Software Requirements	Specification
Processor	Intel i3 or higher	Operating System	Windows 10 or above
RAM	Minimum 4 GB	Frontend Technologies	HTML, CSS
Storage	At least 100 GB free space	Backend Technology	PHP
System Type	64-bit OS	Database	MySQL
Input Devices	Keyboard and Mouse	Server Environment	XAMPP / WAMP
—	—	Web Browser	Chrome / Edge

7. System Design

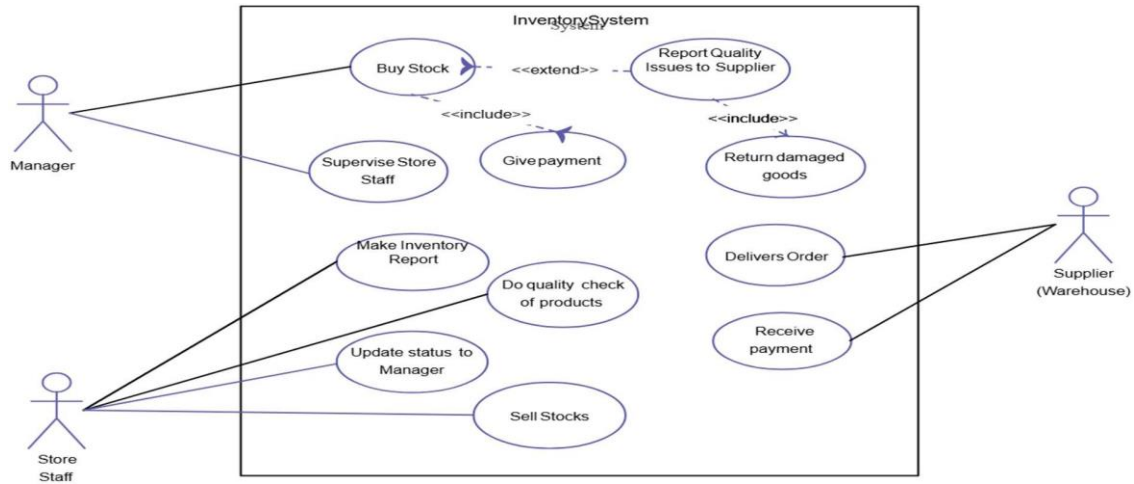
Architecture Diagram



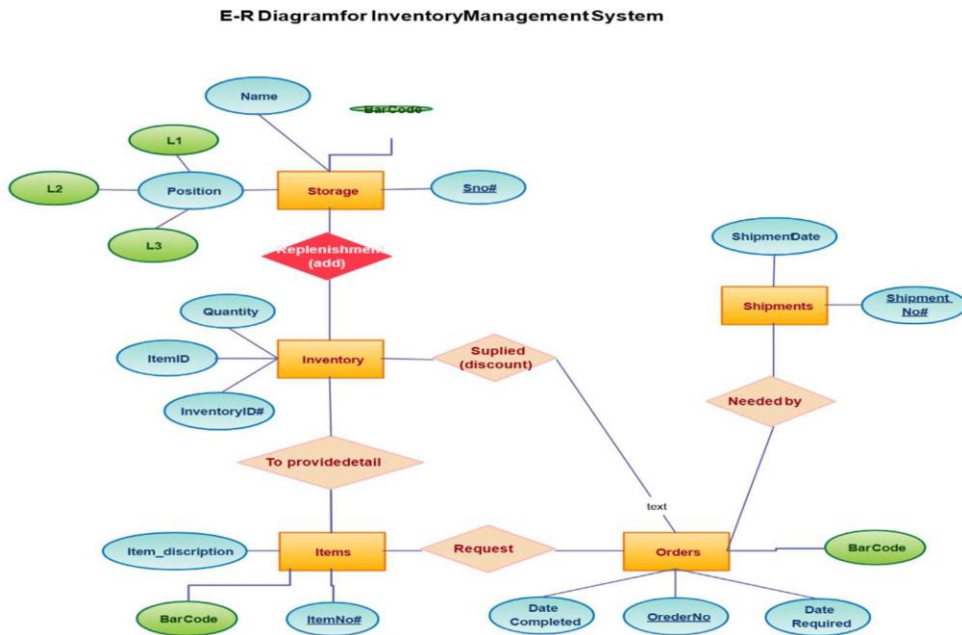
Data Flow Diagram (DFD)



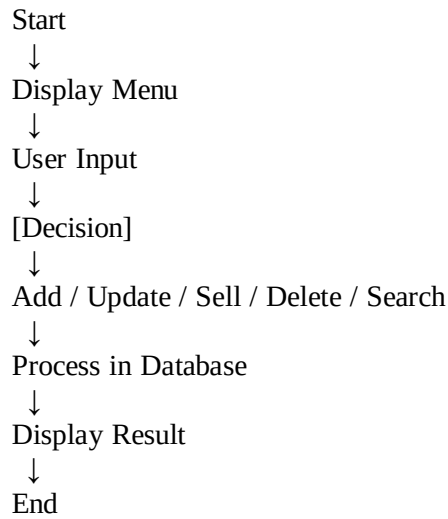
Use case diagram



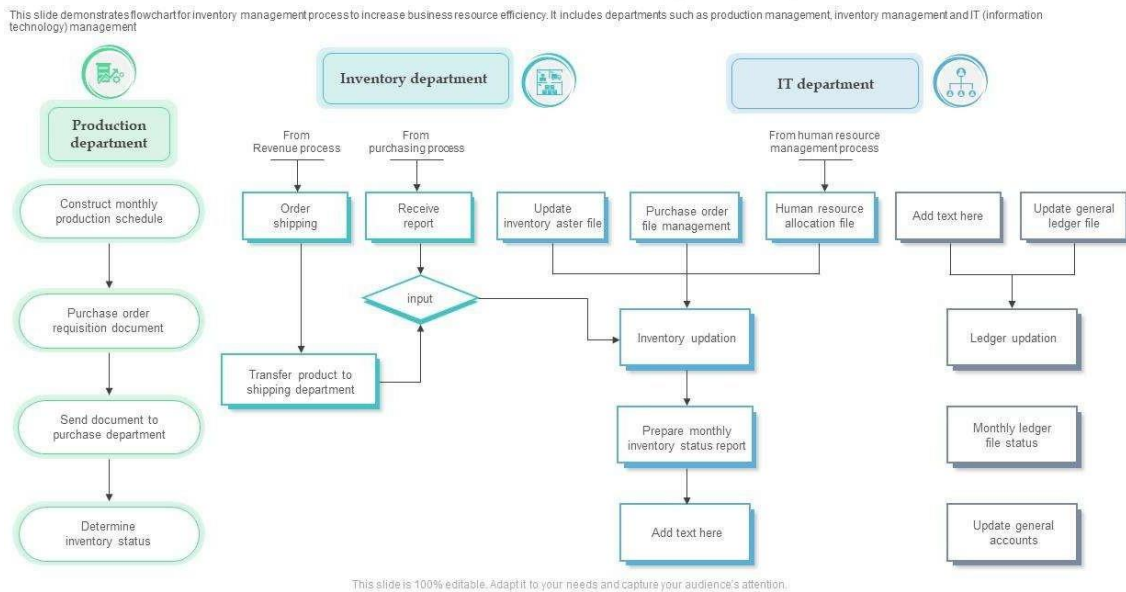
ER diagram



Algorithm/Model flowchart



Inventory management process flow chart :



8. Implementation

Technologies and Tools Used

The Supermarket Inventory Management System is developed using a combination of web technologies and database tools to ensure efficient performance and user interaction. The frontend of the system is designed using HTML and CSS, which provide a simple and interactive user interface. The backend is implemented using PHP, which handles the business logic and communicates with the database. MySQL is used as the database management system to store and manage product data. The system is executed in a local server environment using tools such as XAMPP or WAMP, and it is accessed through web browsers like Google Chrome or Microsoft Edge.

Programming Languages

The project integrates multiple programming languages, each serving a specific role in the system's development.

- **HTML(HyperTextMarkupLanguage):**
Used to design the basic structure of web pages. It defines elements such as forms, tables, buttons, and content layout.
- **CSS(CascadingStyleSheets):**
Used to enhance the visual appearance of the application. It controls styling aspects such as colors, fonts, spacing, and overall layout, making the interface user-friendly and responsive.
- **PHP(HypertextPreprocessor):**
Acts as the server-side scripting language. It processes user inputs, handles form submissions, manages sessions, performs validations, and controls the application logic. PHP also connects the frontend with the database.
- **SQL(StructuredQueryLanguage):**
Used for managing and interacting with the MySQL database. It performs operations such as:
 - INSERT – to add new records
 - UPDATE – to modify existing data
 - DELETE – to remove records
 - SELECT – to retrieve required data

Algorithms or Models Implemented

The system is based on a simple CRUD (Create, Read, Update, Delete) model for managing inventory data. The following logical operations are implemented:

- Add Product Algorithm: Inserts new product details into the database.
- Update Stock Algorithm: Increases the quantity of an existing product.
- Sell Product Algorithm: Checks product availability and reduces quantity by one if stock is available.
- Delete Product Algorithm: Removes a product record from the database.
- Search Algorithm: Retrieves product details based on product ID or name using SQL queries.

Step-by-Step Process / Module Description

1. Database Connection Module

This module establishes a connection between the application and the MySQL database using PHP (db.php). It ensures that all operations can interact with the database successfully.

2. Add Product Module

In this module, the user enters product details such as name, price, and quantity through a form. The data is sent to the backend using the POST method and inserted into the database using an SQL INSERT query.

3. View Products Module

This module retrieves all product records from the database and displays them in a tabular format on the user interface. It uses SQL SELECT queries to fetch data.

4. Update Stock Module

This module allows users to increase the quantity of an existing product. The system updates the product quantity using an SQL UPDATE query.

5. Sell Product Module

In this module, the system checks whether the product is available in stock. If the quantity is greater than zero, it decreases the quantity by one. If not, it displays an “Out of stock” message.

6. Delete Product Module

This module enables the user to remove a product from the inventory. The product is deleted from the database using an SQL DELETE query.

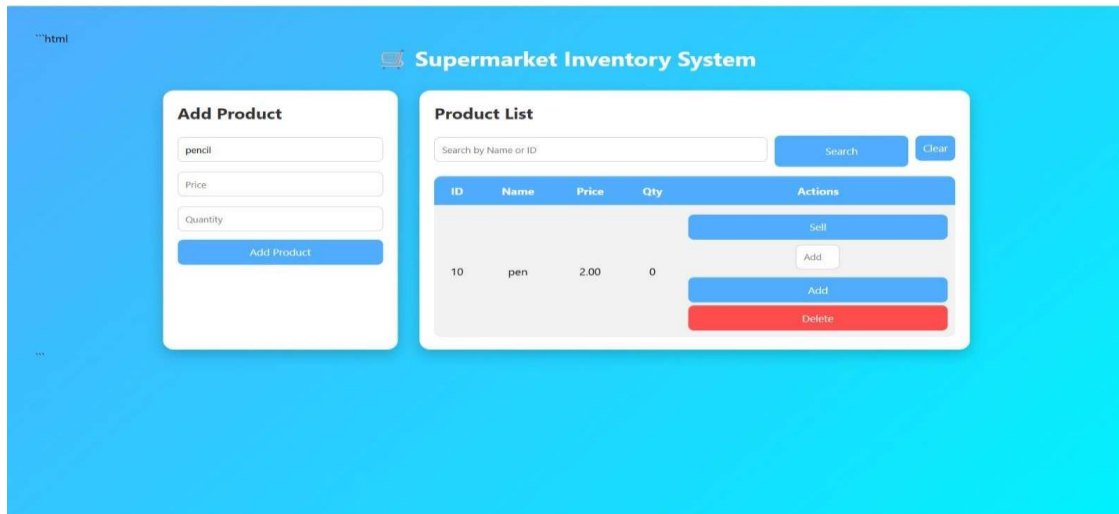
7. Search Module

This module allows users to search for products based on product ID.

9. Results and Discussion

Screenshot of Output

The developed Supermarket Inventory Management System was successfully implemented and tested in a local environment. The system provides an interactive and user-friendly interface for managing product inventory efficiently.



Comparison with existing methods

Comparison with Existing Methods		
Aspect	Existing System (Manual)	Proposed System (Digital)
Data Handling	Manual, error-prone	Automated and accurate
Speed	Slow	Fast
Data Retrieval	Difficult	Instant
Stock Management	Not real-time	Real-time updates
Efficiency	Low	High

Interpretation of Results

The results obtained demonstrate that the Supermarket Inventory Management System operates efficiently and accurately. All core functionalities, including adding products, updating stock, processing sales, deleting records, and searching for items, are executed successfully without errors.

The system provides real-time updates of inventory levels, ensuring that stock records remain accurate and up to date. In comparison to traditional manual methods, it significantly reduces time consumption and minimizes the chances of human error.

Additionally, the system features a user-friendly interface, enabling users to perform operations quickly and with ease. Overall, the results confirm that the system is reliable, efficient, and well-suited for managing inventory in small-scale supermarket environments.

Conclusion and Future Work

Conclusion

The Supermarket Inventory Management System was successfully designed and implemented to manage product records and stock levels efficiently. The system replaces traditional manual methods with a digital solution, enabling accurate and real-time tracking of inventory. All core functionalities such as adding products, updating stock, selling items, deleting records, and searching for products were implemented effectively using PHP and MySQL.

The results demonstrate that the system improves data accuracy, reduces human errors, and minimizes the time required for maintaining inventory records. The user-friendly interface ensures ease of use, making the system suitable even for users with limited technical knowledge. Overall, the project provides a simple, reliable, and efficient solution for small-scale supermarket inventory management.

Key Findings and Contributions

- Successful implementation of CRUD operations for efficient data handling
- Real-time stock updating ensures accurate inventory tracking
- Reduction in manual errors compared to traditional methods
- Improved speed and efficiency in managing product data
- Development of a simple and user-friendly interface

Limitations of the Project

- The system supports only basic inventory operations
- No user authentication or login system is implemented
- Does not include advanced features such as billing or reporting
- Limited to a single product table without supplier or customer management
- Runs on a local server and is not deployed online

Future Work

The system can be further enhanced by adding advanced features to improve functionality and scalability. Future improvements may include:

- Implementation of user authentication and role-based access control
- Integration of billing and invoice generation systems
- Addition of supplier and customer management modules
- Development of graphical reports and analytics for better decision-making
- Deployment of the system on a cloud platform for remote access
- Enhancement of the user interface with modern frameworks

